1.8 Counting the Rationals

WORKSHEET

The question

STEP III 2020, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Before proving anything, compute the first fifteen or so terms. What do you notice about the ratios u_n / u_{n+1}?

STAGE 1 Odd beats even

Ask yourself Write each of these pairs in terms of u_k and u_{k+1}. Which term is bigger?

STAGE 2 Consecutive terms are coprime

Ask yourself Suppose $d > 1$ divides both u_{2k} and u_{2k+1}, that is, both u_k and $u_k + u_{k+1}$. What else does d divide?

STAGE 3 No pair appears twice

Pause and try Stage 1 tells you something useful: if you see a consecutive pair (a, b), you can tell whether a sits at an odd or an even position. How? And can you then recover the pair that came before it, in the sense of Stage 2?

STAGE 4 Every coprime pair appears

Ask yourself Suppose some coprime pair (a, b) never appears consecutively. Which such pair should you look at?

Try this next

1. **Find the fraction.** For which n is $f(n) = \frac{3}{5}$? And $f(n) = \frac{5}{3}$? Use the parent rule to work backwards to $\frac{1}{1}$.

2. **Newman's formula.** Check that Newman's formula produces the first ten fractions of the list, starting from $x_1 = 1$.

3. **Cantor's diagonal argument.** Suppose you had a list of all real numbers between 0 and 1, written as decimals. Construct a decimal that differs from the nth number in the list in its nth digit. Why does this show the list is incomplete? (What care is needed with decimals like $0.0999\ldots = 0.1$?)

4 Countable unions. HARDER Show that the set of all pairs (m, n) of positive integers is countable by giving an explicit bijection with the positive integers. (Try $2^{m-1}(2n-1)$.)

Further reading

- **Neil Calkin & Herbert Wilf, "Recounting the rationals"** (*American Mathematical Monthly*, 2000). Four pages, entirely readable.
- **Aigner & Ziegler, *Proofs from THE BOOK***, Chapter 17, "Sets, functions, and the continuum hypothesis", which includes the Calkin–Wilf tree.
- **Ian Stewart & David Tall, *The Foundations of Mathematics*** (Oxford University Press). Countability, Cantor and the real numbers, as in a first-year course.

The full guided solution, and what the question leads to at university, is in Section 1.8:
https://beyondstep.keithmcnulty.org/ch01/counting-rationals.html